22/7/26

Experiment 7

Sliding Window protocol.

Aim:

To write a program to implement flow control at data link layer using sliding window protocol. Simulate the flow of frames from one node to another.

Code:

```
import time
message = input ("Enter the text message : ")
window_size = int(input ("Enter window size : "))
error_frame = int(input ("Enter frame to err or no error : 0"))
if window_size <= 0 :
    print ("Invalid window size")
    exit ()

print ("\n message :", message)
print ("window size :", window_size)
start = 0
expected_frames = 1
error_induced = False
while start < len(message):
    end = min (start + window_size, len(message))

Sender buffer = [ ]
for i in range (start, end):
    frame_no = i + 1
    data = message [i]
```

```
Sender_buffer.append([frame_no, data])
Print("FRAME_NO =", frame_no, "DATA =", data)
~~print("Sending frames")~~
time.sleep(1)
error_found = False
for frame in Sender_buffer:
    frame_no = frame[0]
    data = frame[1]
if frame_no == error_frame and not error_induced:
    received_frame = frame_no + 1
    error_induced = True
else:
    received_frame = frame_no
Print("Received frame:", receiver_frame, "DATA:", data)
if received_frame == expected_frame:
    print("Frame", received_frame, "is correct")
    expected_frame += 1
else:
    print("Expated frame:", expected_frame)
    print("Received Frame:", received_frame)
    print("NACK =", expected_frame)
    print("NACK received by sender")
    print("Resending frames from frame", expected_frame)
```

```
error_found = True
break
if error_found:
    start = expected_frame - 1
    time.sleep(1)
    continue
if ack == end + 1:
    print("In Ack is correct")
    print("sliding window moves forward")
    start = end
else:
    print("Incorrect Ack")
    print("waiting for correct Ack")
print("transmission completed")
print("Original message:", message)
print("All frames successfully transmitted")
```

Input

```
Enter text message : HELLO
Enter window size : 3
Enter Error frame (0 for no error): 2
```

O/P:

```
message : HELLO
window size : 3
Frame_no = 1    data = H
Frame_no = 2    data = E
```

Frame_no = 3 data = L

Sending frames . . .

Received Frame : 1 Data : H

Frame 1 is correct

Received Frame : 3 Data : E

Error detected :

Expected Frame : 2

Received Frame : 3.

NACK : 2

NACK Received by Sender

Resending Frames from frame 2

Frame_no = 2 data = E

Frame_no : 3 data = L

Frame_no = 4 data = L

Sending frames . . .

Received Frame : 2 data : E

Frame 2 is correct

Received Frame : 3 Data : L

Frame 3 is correct

Received Frame : 4 data : L

Frame 4 is correct

Ack : 5

Ack received by Sender

Expected Ack : 5

Ack is correct

{Sliding window moves forward}

Received Frame : 5 Data : 0

Ack : 6

Ack received by Sender

Expected Ack : 6

Ack is correct

{Sliding window moves forward}

Original Message : HELLO

All frames successfully transmitted

Result:

Program implemented successfully to implement sliding window protocol & simulated flow frames